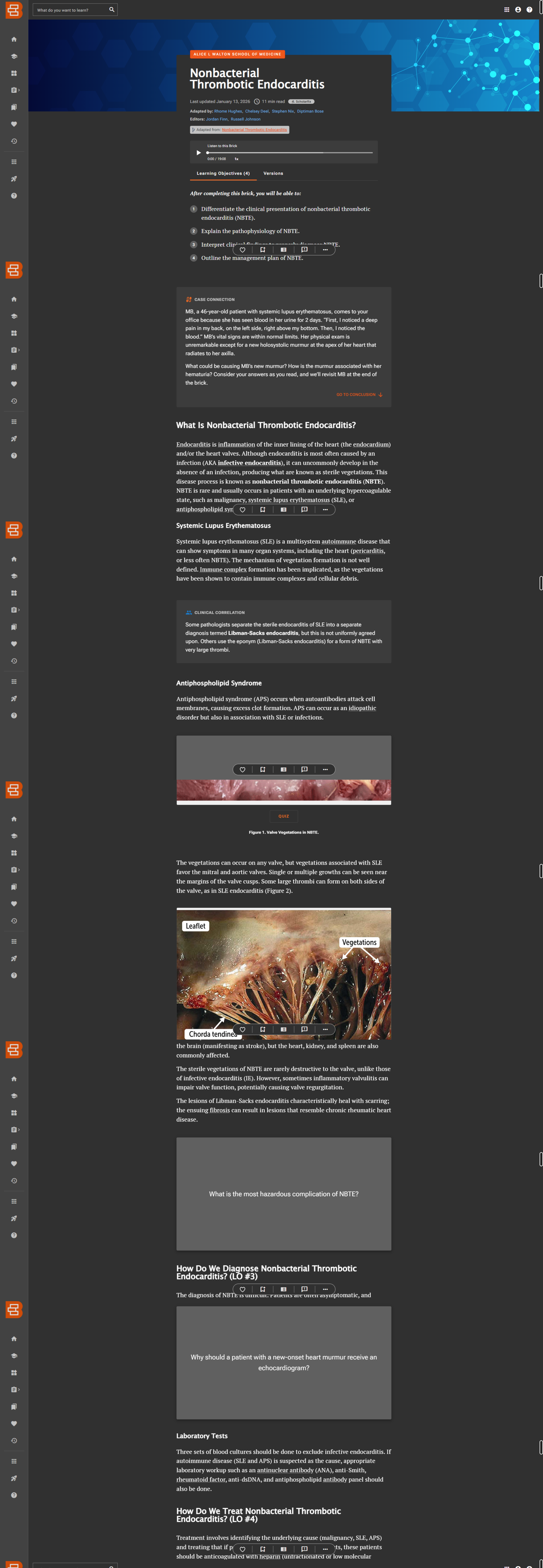

Nonbacterial Thrombotic Endocarditis

Last updated January 13, 2026 · 11 min read · ScholarRx

Adapted by: Rhome Hughes, Chelsey Deel, Stephen Nix, Diptiman Bose

Editors: Jordan Finn, Russell Johnson

Adapted from: Nonbacterial Thrombotic Endocarditis

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of nonbacterial thrombotic endocarditis (NBTE).
2. Explain the pathophysiology of NBTE.
3. Interpret clin [illegible] NBTE.
4. Outline the management plan of NBTE.

CASE CONNECTION

MB, a 46-year-old patient with systemic lupus erythematosus, comes to your office because she has seen blood in her urine for 2 days. "First, I noticed a deep pain in my back, on the left side, right above my bottom. Then, I noticed the blood." MB's vital signs are within normal limits. Her physical exam is unremarkable except for a new holosystolic murmur at the apex of her heart that radiates to her axilla.

What could be causing MB's new murmur? How is the murmur associated with her hematuria? Consider your answers as you read, and we'll revisit MB at the end of the brick.

What Is Nonbacterial Thrombotic Endocarditis?

Endocarditis is inflammation of the inner lining of the heart (the endocardium) and/or the heart valves. Although endocarditis is most often caused by an infection (AKA **infective endocarditis**), it can uncommonly develop in the absence of an infection, producing what are known as sterile vegetations. This disease process is known as **nonbacterial thrombotic endocarditis (NBTE)**. NBTE is rare and usually occurs in patients with an underlying hypercoagulable state, such as malignancy, systemic lupus erythematosus (SLE), or antiphospholipid syn

Systemic Lupus Erythematosus

Systemic lupus erythematosus (SLE) is a multisystem autoimmune disease that can show symptoms in many organ systems, including the heart (pericarditis, or less often NBTE). The mechanism of vegetation formation is not well defined. Immune complex formation has been implicated, as the vegetations have been shown to contain immune complexes and cellular debris.

CLINICAL CORRELATION

Some pathologists separate the sterile endocarditis of SLE into a separate diagnosis termed **Libman-Sacks endocarditis**, but this is not uniformly agreed upon. Others use the eponym (Libman-Sacks endocarditis) for a form of NBTE with very large thrombi.

Antiphospholipid Syndrome

Antiphospholipid syndrome (APS) occurs when autoantibodies attack cell membranes, causing excess clot formation. APS can occur as an idiopathic disorder but also in association with SLE or infections.

Figure 1. Valve Vegetations in NBTE.

The vegetations can occur on any valve, but vegetations associated with SLE favor the mitral and aortic valves. Single or multiple growths can be seen near the margins of the valve cusps. Some large thrombi can form on both sides of the valve, as in SLE endocarditis (Figure 2).

the brain (manifesting as stroke), but the heart, kidney, and spleen are also commonly affected.

The sterile vegetations of NBTE are rarely destructive to the valve, unlike those of infective endocarditis (IE). However, sometimes inflammatory valvulitis can impair valve function, potentially causing valve regurgitation.

The lesions of Libman-Sacks endocarditis characteristically heal with scarring; the ensuing fibrosis can result in lesions that resemble chronic rheumatic heart disease.

What is the most hazardous complication of NBTE?

How Do We Diagnose Nonbacterial Thrombotic Endocarditis? (LO #3)

The diagnosis of NBTE is difficult. Patients are often asymptomatic, and

Why should a patient with a new-onset heart murmur receive an echocardiogram?

Laboratory Tests

Three sets of blood cultures should be done to exclude infective endocarditis. If autoimmune disease (SLE and APS) is suspected as the cause, appropriate laboratory workup such as an antinuclear antibody (ANA), anti-Smith, rheumatoid factor, anti-dsDNA, and antiphospholipid antibody panel should also be done.

How Do We Treat Nonbacterial Thrombotic Endocarditis? (LO #4)

Treatment involves identifying the underlying cause (malignancy, SLE, APS) and treating that if p ts, these patients should be anticoagulated with heparin (unfractionated or low molecular

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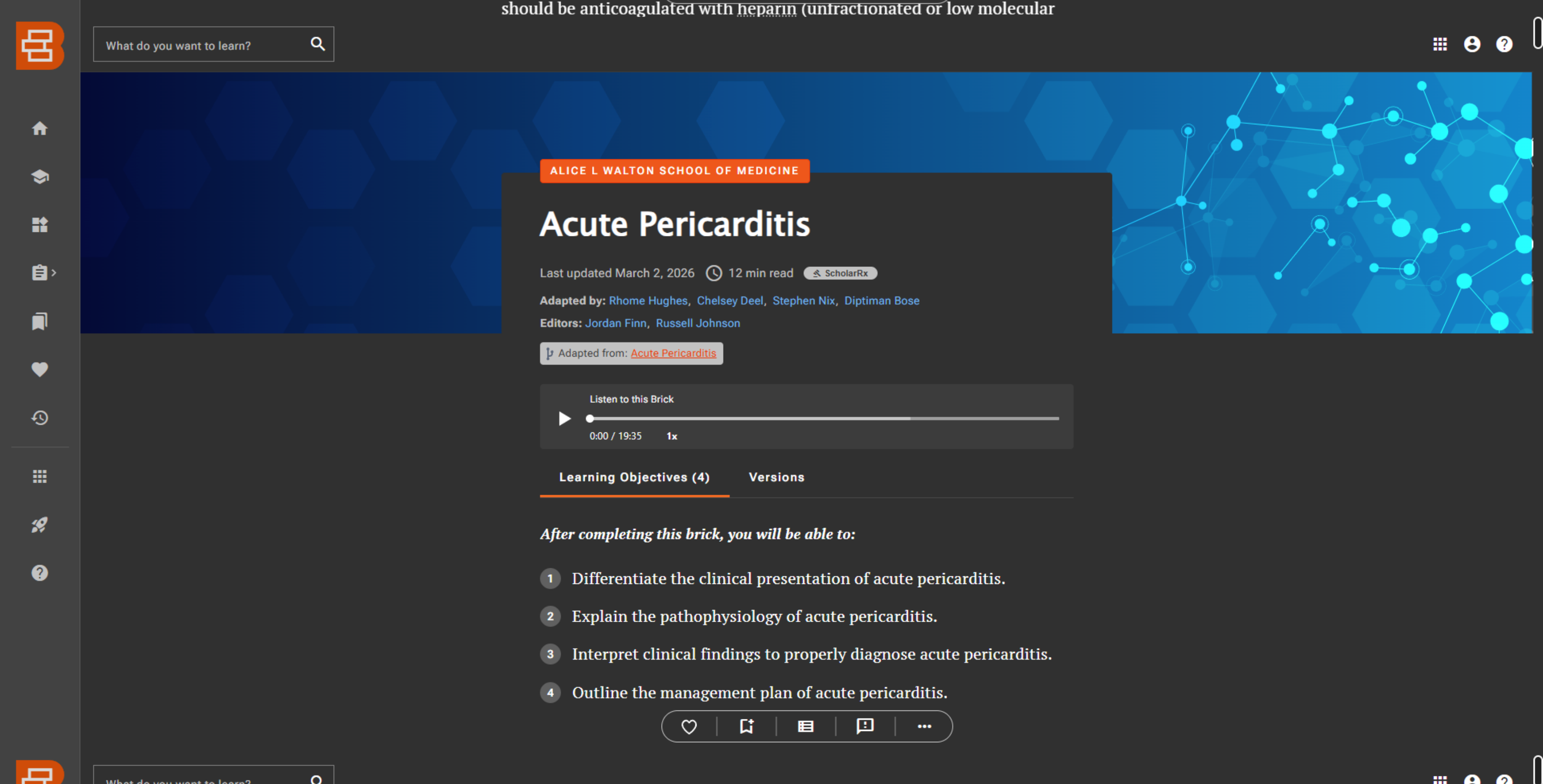


Acute Pericarditis

Last updated March 2, 2026 · 12 min read · ScholarRx

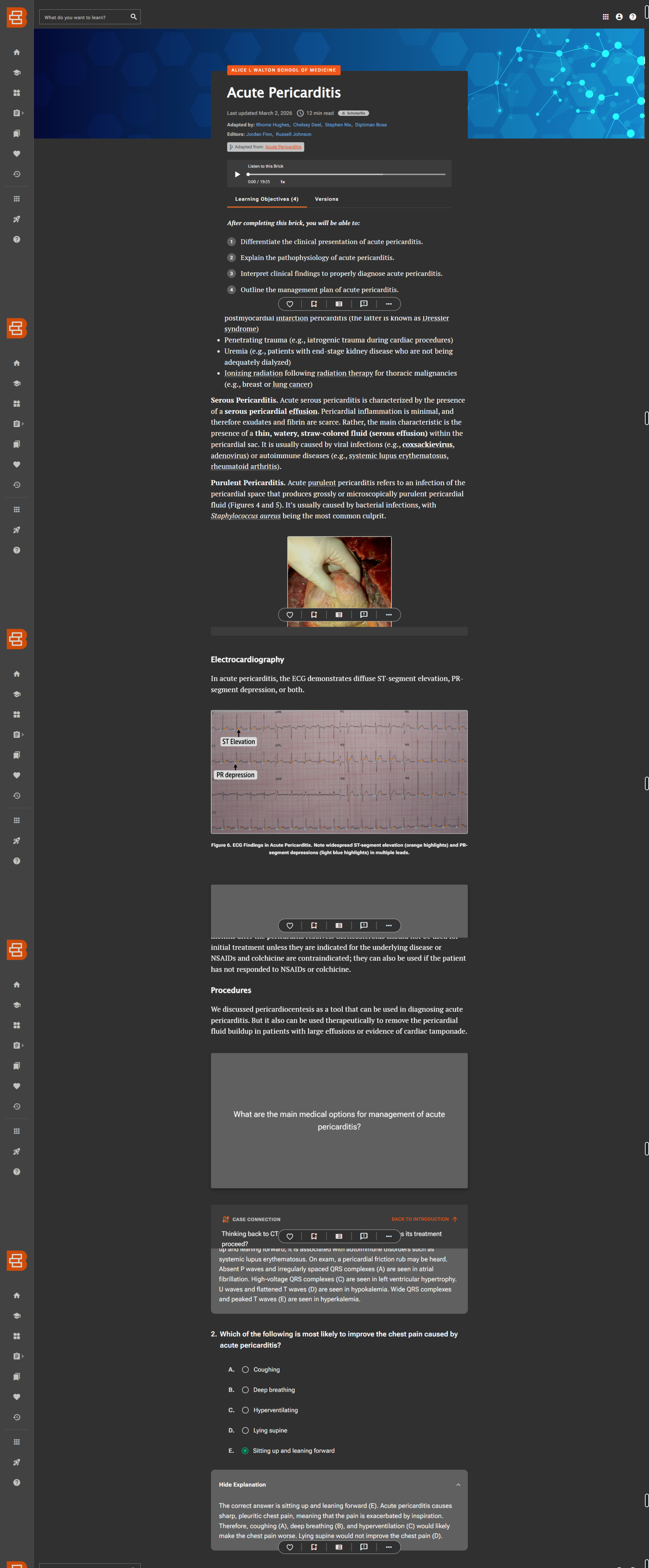
Adapted by: Rhome Hughes, Chelsey Deel, Stephen Nix, Diptiman Bose

Editors: Jordan Finn, Russell Johnson

Adapted from: Acute Pericarditis

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of acute pericarditis.
2. Explain the pathophysiology of acute pericarditis.
3. Interpret clinical findings to properly diagnose acute pericarditis.
4. Outline the management plan of acute pericarditis.



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postmyocardial infarction pericarditis (the latter is known as Dressler syndrome)
- Penetrating trauma (e.g., iatrogenic trauma during cardiac procedures)
- Uremia (e.g., patients with end-stage kidney disease who are not being adequately dialyzed)
- Ionizing radiation following radiation therapy for thoracic malignancies (e.g., breast or lung cancer)

Serous Pericarditis. Acute serous pericarditis is characterized by the presence of a **serous pericardial effusion**. Pericardial inflammation is minimal, and therefore exudates and fibrin are scarce. Rather, the main characteristic is the presence of a **thin, watery, straw-colored fluid (serous effusion)** within the pericardial sac. It is usually caused by viral infections (e.g., **coxsackievirus**, adenovirus) or autoimmune diseases (e.g., systemic lupus erythematosus, rheumatoid arthritis).

Purulent Pericarditis. Acute purulent pericarditis refers to an infection of the pericardial space that produces grossly or microscopically purulent pericardial fluid (Figures 4 and 5). It's usually caused by bacterial infections, with *Staphylococcus aureus* being the most common culprit.

Electrocardiography

In acute pericarditis, the ECG demonstrates diffuse ST-segment elevation, PR-segment depression, or both.

Figure 6. ECG Findings in Acute Pericarditis. Note widespread ST-segment elevation (orange highlights) and PR-segment depressions (light blue highlights) in multiple leads.

initial treatment unless they are indicated for the underlying disease or NSAIDs and colchicine are contraindicated; they can also be used if the patient has not responded to NSAIDs or colchicine.

Procedures

We discussed pericardiocentesis as a tool that can be used in diagnosing acute pericarditis. But it also can be used therapeutically to remove the pericardial fluid buildup in patients with large effusions or evidence of cardiac tamponade.

What are the main medical options for management of acute pericarditis?

CASE CONNECTION

Thinking back to CT s its treatment proceed?

up and leaning forward; it is associated with autoimmune disorders such as systemic lupus erythematosus. On exam, a pericardial friction rub may be heard. Absent P waves and irregularly spaced QRS complexes (A) are seen in atrial fibrillation. High-voltage QRS complexes (C) are seen in left ventricular hypertrophy. U waves and flattened T waves (D) are seen in hypokalemia. Wide QRS complexes and peaked T waves (E) are seen in hyperkalemia.

2. Which of the following is most likely to improve the chest pain caused by acute pericarditis?

A. ○ Coughing

B. ○ Deep breathing

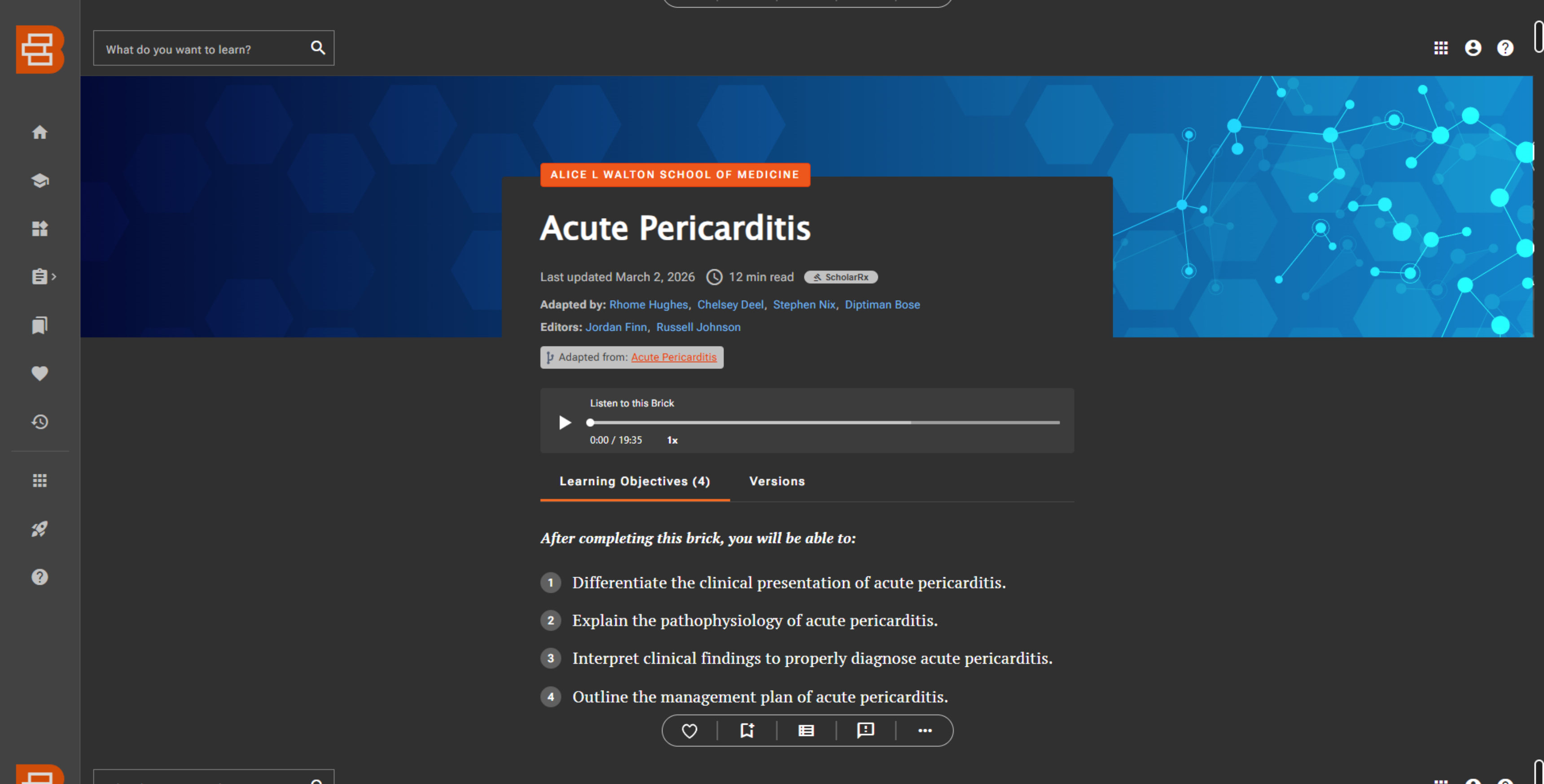
C. ○ Hyperventilating

D. ○ Lying supine

E. ● Sitting up and leaning forward

Hide Explanation

The correct answer is sitting up and leaning forward (E). Acute pericarditis causes sharp, pleuritic chest pain, meaning that the pain is exacerbated by inspiration. Therefore, coughing (A), deep breathing (B), and hyperventilation (C) would likely make the chest pain worse. Lying supine would not improve the chest pain (D).



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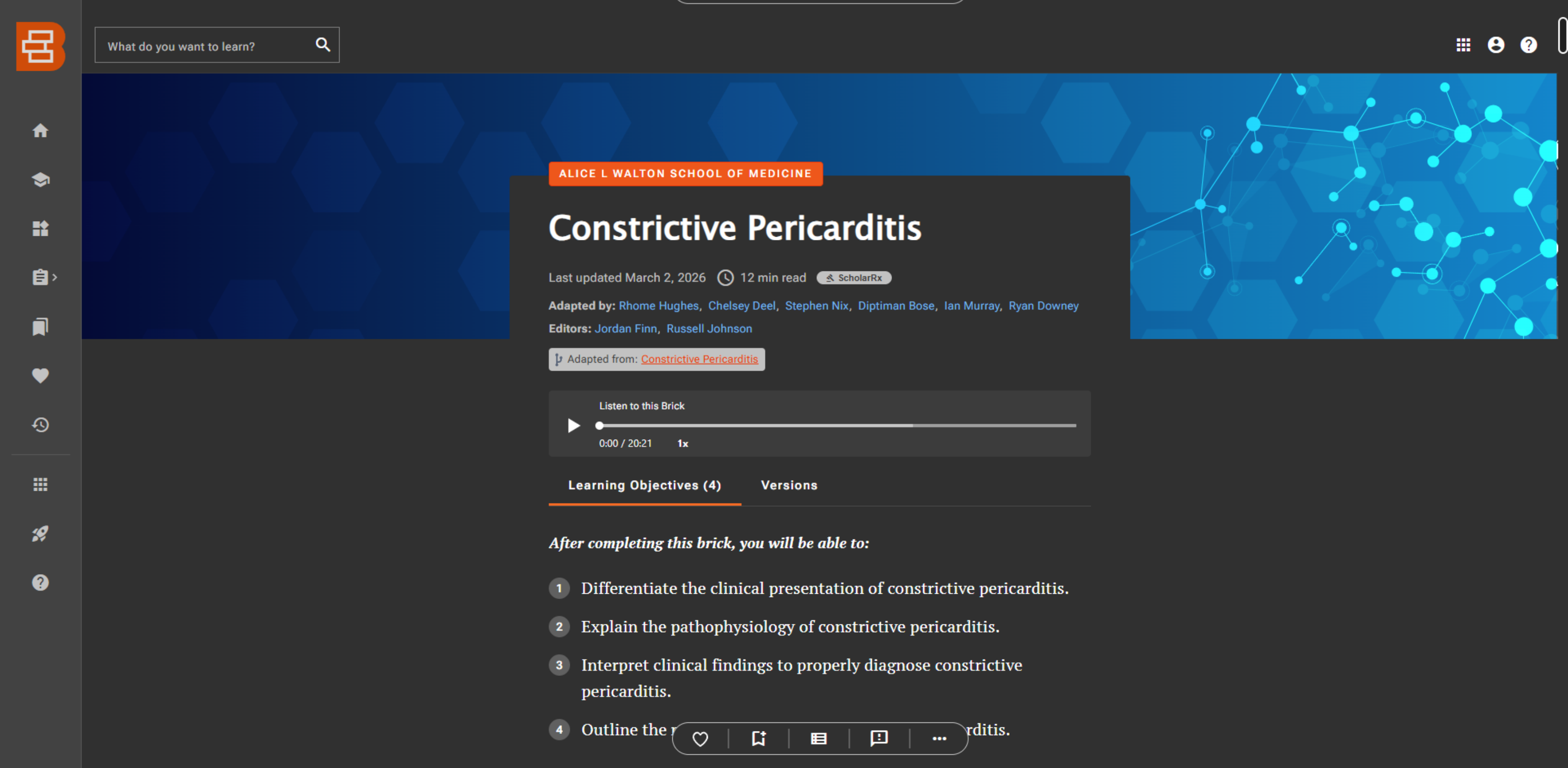

Constrictive Pericarditis

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Adapted by: Rhome Hughes, Chelsey Deel, Stephen Nix, Diptiman Bose, Ian Murray, Ryan Downey

Editors: Jordan Finn, Russell Johnson

Adapted from: Constrictive Pericarditis

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of constrictive pericarditis.
2. Explain the pathophysiology of constrictive pericarditis.
3. Interpret clinical findings to properly diagnose constrictive pericarditis.
4. Outline the r rditis.